

Turbo Equalization with Duobinary Precoding for High-speed Bandwidth-limited IM/DD Transmission Systems

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Abstract We propose turbo equalization with duobinary-BCJR detection to mitigate error floors and boost performance of weak inner codes in bandwidth-limited channels. Simulations and experiments demonstrate significant SNR gains and error floor elimination using duobinary precoding. Analytical limits confirm advantages for high-speed optical links. ©2026 The Author(s)

Introduction

The rapid growth of AI and cloud services is accelerating the evolution of data center networks (DCNs) toward 800 GbE and 1.6 TbE. High-speed short-reach intensity modulation with direct detection (IM/DD) links based on 4-level pulse amplitude modulation (PAM4) are widely adopted due to their cost efficiency and low power consumption. These systems typically employ concatenated forward error correction (CatFEC), consisting of Reed-Solomon (544, 514) and Hamming (128, 120) codes [1–3]. Feed-forward equalization (FFE) combined with post filter (PF) and maximum a posteriori (MAP) detection [4, 5] is commonly used to mitigate inter-symbol interference (ISI) [6]. However, its performance becomes limited in bandwidth-constrained scenarios. Turbo equalization (TE) has recently emerged as a powerful alternative [7–9]. Nevertheless, its combination with duobinary (DB) precoding remains unexplored.

In this work, we investigate TE with DB precoding for bandwidth-limited IM/DD PAM4 systems with a weak soft-decoded inner code. We show that TE significantly improves weak code performance via iterative soft-information exchange, but is insufficient to remove error floors under severe ISI. Experimental results at 112 GBd confirm a 1.5 dB gain and complete elimination of BER floors at the KP4 pre-FEC threshold.

Principle of Turbo Equalization with Duobinary Precoding

Fig. 1 depicts the transmitter and receiver architecture used for TE in this study. The source bit

sequence B is first protected by a Hamming encoder, generating the coded sequence C . To mitigate the impact of burst errors, a pseudo-random interleaver (II) is applied. The interleaved bits are then mapped into PAM4 symbols $b_k \in \{0, 1, 2, 3\}$, which are assumed to be independent and identically distributed (i.i.d.). To prevent the catastrophic error propagation inherent in high-ISI channels, a DB precoder is implemented. The precoded symbol $x'_k \in \{0, 1, 2, 3\}$ is defined recursively by:

$$x'_k = (b_k - x'_{k-1}) \pmod{4}, \quad (1)$$

where x'_{k-1} denotes the previously precoded symbol. These symbols are then linearly mapped to a bipolar constellation $x_k \in \{-3, -1, +1, +3\}$ via the transformation $x_k = 2x'_k - 3$. The generated symbol sequence is transmitted through an ISI channel, the received signal at time index k is expressed as:

$$y_k = x_k + \alpha x_{k-1} + \omega_k, \quad (2)$$

where $\alpha \in [0, 1]$ represents the ISI level and ω_k denotes additive white Gaussian noise (AWGN). This simplified model effectively characterizes the equivalent channel after FFE and noise whitening filtering (NWF), in which controlled and known ISI is introduced [8, 9] and the noise is approximately white.

Fig. 2 illustrates the state transition trellis of the channel model (2) incorporating DB precoding. The DB-BCJR detector computes symbol-wise a posteriori probabilities (APP) metrics in the log domain as

$$\log(P(b_k = b | \mathbf{y})) \approx \max_{i \in \{0, 1, 2, 3\}} \{ \tilde{\rho}(s_k^{[i]}) + \tilde{\gamma}(s_{k-1}^{[i]}, s_k^{[b-i \pmod{4}]}) + \tilde{\beta}(s_k^{[b-i \pmod{4}]}) \}, \quad (3)$$

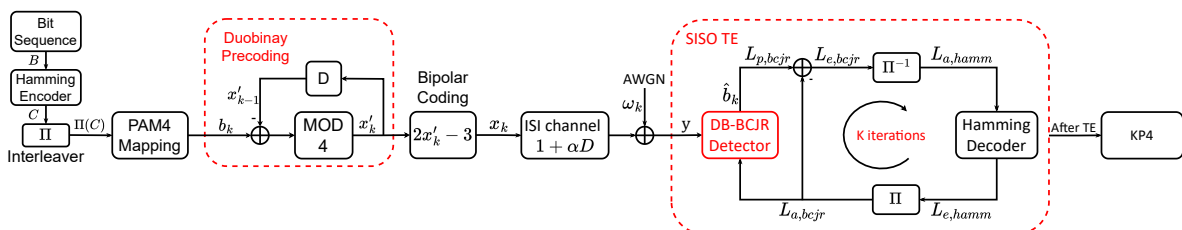


Fig. 1: Block diagram of the simulation system.

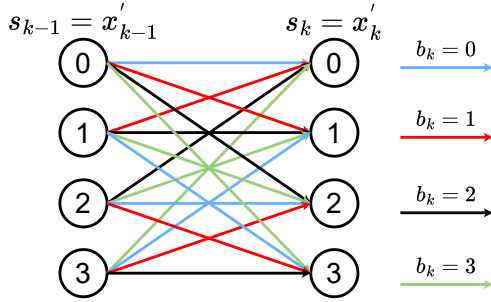


Fig. 2: State transition trellis with DB precoding.

where $\tilde{\rho}$, $\tilde{\gamma}$, and $\tilde{\beta}$ denote the forward state metric, branch metric, and backward state metric, respectively, which are computed following the Max-Log-MAP (MLM) criteria [5].

To summarize, the DB-BCJR detector processes noisy channel observations to generate extrinsic symbol-wise L -values $L_{p,bcjr}$. By subtracting the a priori L -values $L_{a,bcjr}$, the extrinsic information $L_{e,bcjr}$ is obtained. These values are converted into bit-wise L -values and deinterleaved to form the a priori input $L_{a,hamm}$ for the subsequent Hamming decoder. Utilizing the Chase-Pyndiah algorithm [10], the Hamming decoder produces extrinsic bit-wise L -values $L_{e,hamm}$. After interleaving and bit-to-symbol conversion, these values are fed back to the DB-BCJR detector as updated a priori information $L_{a,bcjr}$, thereby completing the TE iterative loop.

Through iterative exchange of soft information between the DB-BCJR detector and the Hamming decoder, the system can potentially approach the performance of joint detection and decoding (JDD).

AIR and BER Analysis

Achievable information rates (AIRs) provide asymptotic performance limits assuming ideal FEC with infinitely long codewords. For the considered channel (2), AIR can be estimated using Monte Carlo sampling combined with a forward sum-product recursion over the joint source-channel trellis [11]. When a BCJR-type detector is employed, the achievable rate is characterized by the generalized mutual information (GMI) [12], which quantifies the performance under a separate de-

tection and decoding (SDD) architecture.

Fig. 3(a) presents the AIRs for four distinct channel configurations. The black curve denotes the i.i.d. PAM4 capacity limit over an AWGN channel. The red curve represents the AIR of the $1 + \alpha D$ channel (2), corresponding to the JDD limit. Meanwhile, the blue and orange curves depict the GMI for an SDD framework using a DB-BCJR detector and a standard BCJR detector applied to the unit-memory channel, respectively. For instance, at $\alpha = 0.8$ and a target code rate of 0.9375 (e.g., a (128, 120) Hamming code), the required SNRs are 13.1 dB for the AWGN baseline and 14.22 dB for JDD. For the SDD schemes, the required SNRs are 15 dB with DB precoding and 15.39 dB without, highlighting a clear SNR advantage provided by DB precoding.

The required SNR values across the four scenarios for $\alpha \in [0, 1]$ at a rate of 0.9375 are summarized in Table 1. It is observed that DB precoding

Tab. 1: Required SNR [dB] at rate = 0.9375

Channel	JDD	SDD (w.o. DB)	SDD (with DB)
$\alpha = 0.0$	13.10	13.10	13.10
$\alpha = 0.1$	13.13	13.15	14.22
$\alpha = 0.2$	13.19	13.25	14.25
$\alpha = 0.3$	13.26	13.44	14.28
$\alpha = 0.4$	13.43	13.71	14.39
$\alpha = 0.5$	13.60	14.06	14.50
$\alpha = 0.6$	13.80	14.48	14.67
$\alpha = 0.7$	14.03	14.95	14.86
$\alpha = 0.8$	14.22	15.39	15.00
$\alpha = 0.9$	14.38	15.74	15.13
$\alpha = 1.0$	14.40	15.88	15.19

begins to outperform the non-precoded scheme when $\alpha \geq 0.7$. The theoretical TE gain can thus be quantified as the SNR gap between the SDD and JDD limits.

For $\alpha = 0.8$, the predicted TE gains are $\Delta_1 = 1.17$ dB without DB and $\Delta_2 = 0.78$ dB with DB. These predictions are consistent with prior results obtained using strong LDPC codes, confirming that GMI accurately predicts TE performance for capacity-approaching codes.

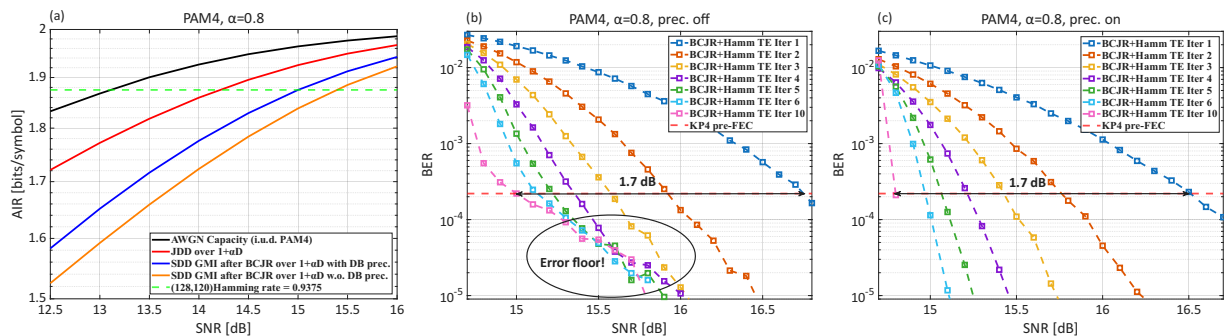


Fig. 3: Performance evaluation of PAM4 signals over the simulated ISI channel (2) ($\alpha = 0.8$): (a) AIR vs. SNR; (b) BER vs. SNR without DB precoding; and (c) BER vs. SNR with DB precoding.

In contrast, as illustrated by the simulation results in Fig. 3(b) and (c), the observed TE gain for the weaker Hamming code reaches 1.7 dB. This exceeds the theoretical prediction, which can be attributed to additional iterative gains arising from the exchange of soft information between the BCJR detector and the weak code. In particular, when the extrinsic information transfer (EXIT) characteristics of the two modules are matched, the combined iterative receiver can approach capacity-approaching performance, as discussed in [13]. Notably, DB precoding not only eliminates the error floor but also further reduces the required SNR by an additional 0.2 dB.

Experimental Setup and Results

Fig. 4 illustrates the experimental setup for a bandwidth-limited 224 Gbps IM/DD PAM4 system.

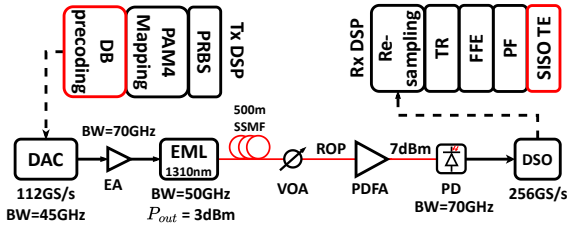


Fig. 4: Experimental setup for the 112GBd PAM4 system.

At the transmitter, a 112 GSa/s digital-to-analog converter (DAC) generates the DB-precoded PAM4 signal, which is amplified by an electrical amplifier (EA) to drive an O-band electro-absorption modulated laser (EML) (50 GHz bandwidth). To compensate for the lack of a receiver-side electrical amplifier, the signal is attenuated by a variable optical attenuator (VOA) and subsequently boosted by a praseodymium-doped fiber amplifier (PDFA). At the receiver, the captured signal undergoes resampling and timing recovery to synchronize the clock and optimize the sampling phase. For offline DSP, 5×10^6 PAM4 symbols are processed, starting with downsampling to 1 sps. Equalization is performed using a 41-tap linear FFE, followed by a 2-tap NWF ($1 + \alpha D$, $\alpha = 0.6$). Finally, SISO TE is implemented.

Figures 5(a) and (b) present the BER versus the ROP without and with DB precoding, respectively. It can be observed that, in the case without precoding, an error floor appears after four TE iterations, indicating that further TE iterations only lead to oscillations around a fixed BER level. In contrast, when DB precoding is applied, the error floor is effectively eliminated, and the BER continues to decrease with additional TE iterations, demonstrating the advantage of the proposed scheme. The TE gain is approximately 1.5 dB for both cases. Since the coefficient of the PF is 0.6, the bandwidth limitation of the experimental setup does not constitute a bottleneck. Nevertheless, duobinary encoding proves very beneficial due to the

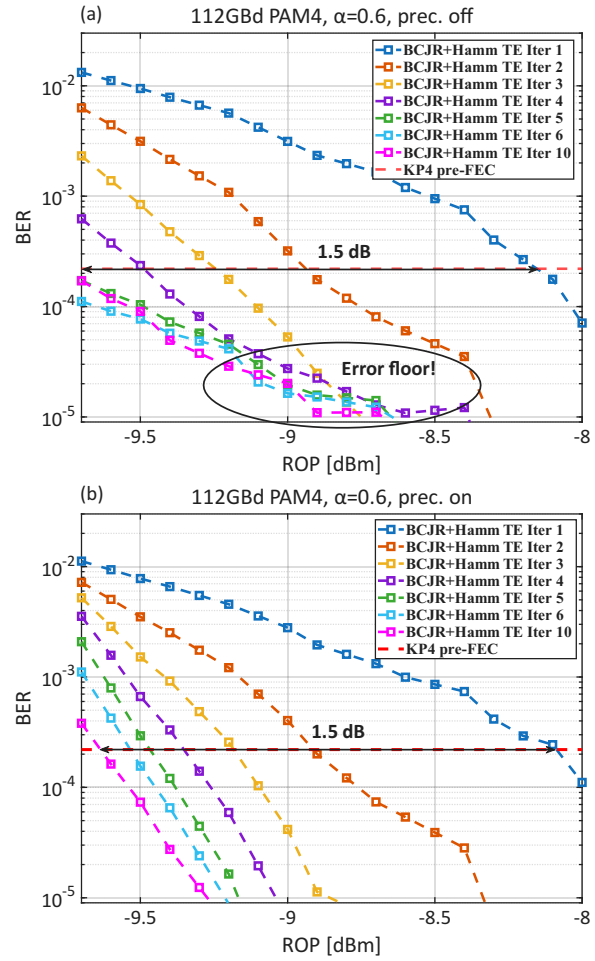


Fig. 5: (a) BER versus ROP without DB precoding; (b) BER versus ROP with DB precoding

weakness of the Hamming code. As a result, the required ROP after TE without DB precoding at the KP4 pre-FEC threshold is slightly lower than that with DB precoding. This observation is also consistent with the results summarized in Table 1.

Conclusions

We have proposed the combination of turbo equalization (TE) and duobinary (DB) precoding for bandwidth-limited IM/DD PAM4 systems with a weak soft-decoded inner code. Our analytical results based on AIRs demonstrate that DB precoding provides a superior GMI baseline for severely impaired ISI channels ($\alpha \geq 0.7$), effectively reducing the required SNR compared to non-precoded schemes at a code rate of 0.9375. Experimental validation in a 224Gbps system confirms that the proposed DB-BCJR-TE architecture not only achieves a significant TE gain of 1.5 dB but also successfully eliminates the error floor observed in non-precoded systems after four iterations. Furthermore, we show that while GMI accurately predicts the performance of strong capacity-approaching codes, TE provides even more pronounced gains for standardized Hamming codes due to additional iterative coding benefits.

References

- [1] IEEE P802.3dj Task Force. "200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Ethernet Task Force", Accessed: Sep. 14, 2025. [Online]. Available: <http://www.ieee802.org/3/dj/>
- [2] K. Zhong et al., "Experimental study of pam-4, cap-16, and dmt for 100 gb/s short reach optical transmission systems", *Optics express*, vol. 23, no. 2, pp. 1176–1189, 2015.
- [3] S. Yin, X. Zhou, and C. Lam. "FEC requirements for 800GbE/1.6TbE optics—An operator's perspective", Accessed: Sep. 14, 2025. [Online]. Available: https://www.ieee802.org/3/df/public/22_07/yin_3df_01_2207.pdf
- [4] L. Bahl, J. Cocke, F. Jelinek, and J. Raviv, "Optimal decoding of linear codes for minimizing symbol error rate (corresp.)", *IEEE Transactions on information theory*, vol. 20, no. 2, pp. 284–287, 1974. DOI: 10.1109/TIT.1974.1055186
- [5] P. Robertson, E. Villebrun, and P. Hoeher, "A comparison of optimal and sub-optimal map decoding algorithms operating in the log domain", in *Proceedings IEEE international conference on communications ICC'95*, IEEE, vol. 2, 1995, pp. 1009–1013.
- [6] T. Wettlin, S. Calabrò, T. Rahman, J. Wei, N. Stojanovic, and S. Pachnicke, "Dsp for high-speed short-reach im/dd systems using pam", *Journal of Lightwave Technology*, vol. 38, no. 24, pp. 6771–6778, 2020.
- [7] S. Chen et al., "Simplified turbo equalizer for bandwidth-limited 50g passive optical networks", *Optics express*, vol. 33, no. 16, pp. 33 169–33 177, 2025.
- [8] T. Kong et al., "Turbo equalization for high-speed pam4 bandwidth-limited im/dd transmission system", in *2025 European Conference on Optical Communications (ECOC)*, IEEE, 2025, pp. 1–4. DOI: 10.1109/ECOC66593.2025.11263167
- [9] T. Kong et al., "High-speed turbo equalization for 224 gb/s pam4 im/dd systems with standardized fec", in *2026 Optical Fiber Communication Conference (OFC)*, Los Angeles, USA, Mar. 2026.
- [10] R. M. Pyndiah, "Near-optimum decoding of product codes: Block turbo codes", *IEEE Transactions on communications*, vol. 46, no. 8, pp. 1003–1010, 2002.
- [11] D.-M. Arnold, H.-A. Loeliger, P. O. Vontobel, A. Kavcic, and W. Zeng, "Simulation-based computation of information rates for channels with memory", *IEEE Transactions on information theory*, vol. 52, no. 8, pp. 3498–3508, 2006.
- [12] G. Kaplan and S. Shamai, "Information rates and error exponents of compound channels with application to antipodal signaling in a fading environment", *AEU. Archiv für Elektronik und Übertragungstechnik*, vol. 47, no. 4, pp. 228–239, 1993.
- [13] A. Ashikhmin, G. Kramer, and S. ten Brink, "Extrinsic information transfer functions: Model and erasure channel properties", *IEEE Transactions on Information Theory*, vol. 50, no. 11, pp. 2657–2673, 2004.